# EXTRA NOTES

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## COMMANDS

1. lsusb-------to check the usb related information
2. lscpu-------to check the cpu related information
3. lspci-------to check info about peripheral component interconnect
4. dmidecode---to check the hardware,motherboard and bios related info
5. tty----------terminal type show
6. w-----------to check the load average
7. diff--------to differentiat the files
8. df----------to display diskspace
9. shred ----------> used to securely delete files so that data cannot be recovered

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## USE OF SCP & RSYNC COMMAND

### SCP (Secure Copy Protocol)

Share files from local to server & server to local

1. **Local To Server**
   scp -i ubuntu.pem file ubuntu@<public_ip>:/home/ubuntu
2. **Server To Local**
   scp -i ubuntu.pem ubuntu@<public_ip>:/home/ubuntu/file /root/

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### RSYNC

Share files from local to server & server to local

1. **Local to Server**
   rsync -avz "ssh -i ubuntu.pem" file ubuntu@<public_ip>:/home/ubuntu
2. **server to local**
   rsync -avz "ssh -i ubuntu.pem" ubuntu@<public_ip>:/home/ubuntu/file /root/

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## AWK COMMAND

*awk → Process and extract columns from text*

1. **Print entire file:**

awk '{print}' file.txt

2. **Print first column:**
   awk '{print $1}' file.txt
3. **Print first & third column:**
   awk '{print $1, $3}' file.txt
4. **Print lines where column 3 > 100:**
   awk '$3 > 100 {print $0}' file.txt
5. **Use with delimiter (e.g., CSV with commas):**
   awk -F, '{print $1,$2}' data.csv

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## SED COMMAND

*sed → Stream editor (find & replace, delete, modify text)*

1. **Replace first occurrence of "cat" with "dog":**
   sed 's/cat/dog/' file.txt
2. **Replace all occurrences:**
   sed 's/cat/dog/g' file.txt
3. **Replace in file (in-place):**
   sed -i 's/cat/dog/g' file.txt
4. **Delete line 5:**
   sed '5d' file.txt
5. **Delete lines 2 to 4:**
   sed '2,4d' file.txt

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## Boot Process

it is a process which is started when we power on the computer or the system and it involves loading and initializing the operating system and the other software necessary for the computer to function properly.

System Power on ----------> gives power to the motherboard ----------> started the BIOS ----------> Performs POST ----------> Boot Loader ----------> Searches the OS - Hardware/Pendrive ----------> load on the ram ----------> cpu started ----------> instructions executed

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## SELINUX

1.it’s a advance security layer to enhance the security.
2.SeLinux (Security-Enhanced Linux) is a security feature in Linux that controls what users and programs can do, adding an extra layer of protection to the system.
3.It’s like a strict security guard inside Linux that decides who can access what.

**Selinux modes:**

1. Permissive (0)
2. Enforcing (1)
3. Disabled

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**MIGRATION**

migration is the technique of moving a virtual machine from one host to another host or from one datastore to another datastore.

**Types of migration**

1. cold migration
2. vmotion (Live Migration)
3. suspended migration
4. physical to virtual (P2V)
5. virtual to virtual (V2V)

**Cold migration**

1. The virtual machine moves from one host to another host in powered-off state.
2. So the chances of failure in the cold migration is less than the other migration technique.

**Suspended migration**

1. migrating the virtual machine that is in suspended state/paused state.
2. VM can resume from the same point later.

**Vmotion**

1. Migrates a VM that is powered on.
2. it is also known as live migration.
3. No downtime for the VM.

**Physical to Virtual (P2V)**

1. Converts a physical machine into a virtual machine.

**Virtual to Virtual (V2V)**

2. Similar to P2V, but the source machine is already virtual.

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**WEB SERVER**

1. **monolithic web server:** single server, single module
2. **microservice web server:** different servers, different modules
3. **extended web server:** single server, modules are different

**Example:**

1.mkdir /var/www/html/mobile
2.vim /var/www/html/mobile/index.html
<h1> this is my mobile page </h1>

3.systemctl restart httpd
4.curl localhost/mobile/
<h1> this is my mobile page </h1

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**USES OF PING COMMAND**

1) ping www.google.com --------> to check the connectivity
2) ping -c 3 www.google.com ---> to send only 3 packets
3) ping -i 2 www.google.com ---> 2 sec intervals to send packets
4) ping -c 5 -q www.google.com -------> only give you the summary
5) ping -f www.google.com ------> to send packets as fast as possible to test the network performance
6) ping -s 500 www.google.com ---> change size of the packets
7) ping -w 10 www.google.com ----> to stop printing after 10 sec
8) ping -D www.google.com -------> also print timestamp to remember the time
9) ping -a www.google.com -------> audible

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**USES OF GREP COMMAND**

**vim sample.txt**

1. The quick brown fox jumps over the lazy dog.
2. Linux is a powerful operating system.
3. Grep is used for searching text in files.
4. Sed is used for text manipulation.
5. Awk is great for pattern scanning and reporting.
6. The dog chased the cat.
7. Cats are very clever animals.
8. Shell scripting makes tasks automated.
9. The word "grep" comes from "Global Regular Expression Print".
10. Some people love cats, while others love dogs.

**Find lines containing a word**
grep "dog" sample.txt

**Case-insensitive search**
grep -i "linux" sample.txt

**Show line numbers with match**
grep -n "cat" sample.txt

**Search for whole word only (not part of another word)**
grep -w "cat" sample.txt

**Search for multiple words (OR condition)**
grep -E "cat|dog" sample.txt

**Invert match (show lines that do NOT contain the word)**
grep -v "dog" sample.txt

**Count the number of matches**
grep -c "cat" sample.txt

**Show only the matched word, not the whole line**
grep -o "dog" sample.txt

**Search using regex (lines starting with number 1)**

grep "^1" sample.txt

**Search for lines ending with "dog."**
grep "dog\.$" sample.txt

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## FTP (File Transfer Protocol)

it is a file transfer protocol in which we can communicate with each other in network and shared files within a network.

**port 20 and 21**
1.port 20: transfer the data via data channel.
2.port 21: to establish connection between two computers.

vsftpd ----------------> very secure file transfer protocol daemon
anonymous user---> ftp
conf file--------------> /etc/vsftpd/vsftpd.conf

it is used as one of the most common means of copying file between computers over the internet.

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## SAMBA

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it basically supports the cross platform and its a file server that enables the file sharing across different operating system over a network

conf file------>  /etc/samba/smb.conf
smb-----------> (445) authentification and data transfer
nmb-----------> (139) network, interface NetBIOS system

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## NFS (Network File System)

1. it is a network file system that allows the system to share the directories and files with others over the network.
2. it is a mechanism for storing files on a network

3. Package name ===> nfs-utils, rpc-bind, mountd

4. port number ===> 2049
5. configuration ===> /etc/sysconfig/nfs
6. service user ===> nfsnobody

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**DIFFERENCE BETWEEN THE SAMBA AND THE NFS**

| Samba | NFS |
|---|---|
| 1.it supports the cross-platform | 1.it does not support the cross-platform |
| 2.integrate with its own authentication | 2. integrate with the authentication services like Kerberos, LDAP |
| 3.it has two service ( smb and nmb) | 3.it has also three service ( nfs, mountd, rpc-bind ) |
| 4.port number ( smb-445 and nmb-139 ) | 4. port number for nfs 2049 |
| 5.Provides file & printer sharing. | 5.Provides fast file sharing only (no printer sharing). |
| 6.Slightly slower (more overhead). | 6.Faster on Linux/Unix due to lightweight protocol. |

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**EXPLORE ISCSI**

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**EXPLORE RAID**

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## DHCP (Dynamic Host Configuration Protocol)

1. it automatically assigns the IPs to the client
2. it works on the DORA process.
3. its based on client-server model
4. DHCP is an Application Layer protocol.
5. dhcp ip address range is called scope
6. It works over UDP ports 67 (server) & 68 (client).

**DHCP SERVER CAN PROVIDE**

1. IP address
2. subnet mask
3. domain name
4. default gateway
5. DNS server address
6. Wins server address

**IT WORKS ON THE DORA'S PROCESS**

**Discover**
The client broadcasts a request asking “Is there any DHCP server available to give me an IP?”
**Offer**
The DHCP server replies with an offer (IP address + subnet mask + gateway + DNS info).
**Request**
The client sends back a request saying “I want to use this IP, please confirm.”
**Acknowledge**
The DHCP server confirms and leases the IP to the client.

Configuration file --------> /etc/dhcp/dhcpd.conf

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## DNS (Domain Name System)

1. it can translate the domain name into the ip addresses. so computers can locate and communicate with each other on the internet.
2. Without DNS, we have to remember numbers instead of names to open websites.
3. it works on the application layer.
4. top level domains are-->(.com,.org,.biz,.edu, .net,.mil)
5. it works on the port number tcp/udp 53

1.resolve name to ips ---------> forward lookup
2.resolve ips to name ---------> reverse lookup

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## NTP (NETWORK TIME PROTOCOL)

Used to synchronize the system clock with a remote time server over a network.

1. package ====> chrony -y
2. port no =====> 123/udp
3. config ======> /etc/chrony.conf
4. service =====> chronyd

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**EXPLORE SSL & TLS**

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**THREE WAYS TO CREATE A USER**

**USING USERADD**

useradd username

**USING ADDUSER**

adduser username

**USING MANUAL METHOD**

1. useradd username
2. mkdir /home/username
3. chown username:username /home/username
4. passwd username

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**SCENARIO BASED QUESTIONS ON THE LINUX**

Scenario: Copy all files owned by user
Question: Copy all files owned by mayur to /backup without changing ownership.

====> find / -user mayur -exec cp -aprvf '{}' /backup \;

==========

Scenario: Cron job not running

====>>
1.check the service

systemctl status crond

2.verify syntax

crontab -l

3.check the logs

/var/log/cron

==========

Scenario: Recover deleted command

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=====>>> 1.yum provides head
2.yum reinstall coreutils

==========

Scenario: Create compressed backup

====>>> tar -cJvf /backup/etc.tar.xz /etc

==========

Scenario: Extend LVM

1.pvcreate /dev/sdb2
2.vgextend vg /dev/sdb2
3.lvextend -L +1G /dev/vg/lv
4.e2fsck -f /dev/vg/lv
5.resize2fs /dev/vg/lv

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Scenario: Reduce LVM

1.e2fsck -f /dev/vg/lv
2.resize2fs /dev/vg/lv
3.lvreduce -L -550M /dev/vg/lv

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Scenario: Convert partition to swap

1.fdisk /dev/sdc  # change type to 82
2.partprobe
3.mkswap /dev/sdc1
4.swapon /dev/sdc1
5.swapoff /dev/sdc1

Scenario: SSH issue (machine should not be connected)

1.check ping command
2.check the service---> systemctl status sshd
3.check the port number 22
4.check the firewall--> firewall-cmd --list-all

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Scenario: Sudo without password

vim /etc/sudoers

linux   ALL=(ALL)   NOPASSWD:/sbin/dmidecode

==========

Scenario: User home directory not created

=====> when create the user use -m to create the home directory ( useradd -m username)

==========

Scenario: You want to set password expiry for all users

====>>>

Set default in /etc/login.defs:

PASS_MAX_DAYS 30

==========

Scenario: User needs temporary root

=====> add user into the /etc/sudoers file

mayur   ALL=(ALL)     ALL

==========

Scenario: You need to find which process is using port 8080

ss -tlnp | grep 8080

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Scenario: Need to mount ISO file

mount -o loop /root/centos.iso /mnt/iso

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Scenario: Two users need different shells

usermod -s /bin/bash raj
usermod -s /sbin/nologin om

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What is inode number?

Unique index pointing to file metadata.

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How do you check current load on system?

uptime
top

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## LINUX IMPORTANT COMMANDS

**Navigation & Basic Commands**
1.pwd
2.ls -al
3.cd
4.touch
5.rm -rf
6.mkdir

**File Read & Edit**
1.cat
2.tail
3.head
4.less
5.more
6.vim
7.nano

**Search & Filter**
1.grep
2.find
3.locate

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4.awk
5.sed

**Permissions**
1.chmod
2.chown
3.chgrp

**Monitoring**
1.top
2.free -h
3.df -Th
4.du -sh

**Networking**
1.ifconfig
2.ip a
3.ip addr
4.curl
5.wget
6.curl -O

**Processes**
1.ps -aux
2.kill -9

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